

1 Logic: propositions, truth tables and classifications

Scope. The two complete units visible in your syllabus: **Logic** and **Combinatorics**. The previous unit is cropped in the photo and is not reconstructed here. This guide has **20 numbered worked questions plus focused method extensions**, compact theory, and black-and-white diagrams. The methods follow your linked Unit 2 and Unit 3 notes, condensed for revision. Each numbered section is one page.

Method alignment. Unit 2: truth assignments, row/column quantifier tests, justified proof steps, and answer substitutions. Unit 3: auxiliary equation, CF + PI, worst-case pigeonholes, shifted series, and partial fractions. Source links: **Unit 2 notes**; **Unit 3 notes**. Examples and explanations are condensed and adapted; the full practice banks are not reproduced.

A **proposition** is a declarative statement with one truth value, true (T) or false (F). “7 is prime” is a proposition; “close the door” is not. “ $x > 3$ ” is an open statement until x is assigned or quantified. Let p, q, r denote propositions.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

$\neg$ means NOT; $\wedge$ means AND; $\vee$ means inclusive OR (one or both); $\rightarrow$ means implication; $\leftrightarrow$ means “if and only if”. An implication is false *only* when its hypothesis is true and its conclusion false. For k distinct proposition letters, a complete table has 2^k rows. Use parentheses to make the order of operations explicit.

Term	Precise meaning
Tautology / valid formula	True under every truth assignment.
Contradiction / unsatisfiable	False under every truth assignment.
Satisfiable	True under at least one assignment (includes tautologies).
Contingency	True under some assignments and false under others.
Equivalence	$P \equiv Q$ iff P and Q have identical truth values on every row.
Valid argument	No assignment makes all premises true and the conclusion false.

Here P, Q may be compound formulas. $\equiv$ asserts equivalence; $\leftrightarrow$ is a connective.

Worked question 1: Classify a formula and test an argument

Question. Classify $F = [(p \rightarrow q) \wedge p] \rightarrow q$. Is the argument $p \rightarrow q, p \therefore q$ valid?

Formula / rule first. $A \rightarrow B \equiv \neg A \vee B$; an argument $P_1, \dots, P_m \therefore C$ is valid iff $(P_1 \wedge \dots \wedge P_m) \rightarrow C$ is a tautology.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	F
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

All entries in the last column are T: F is a **tautology**, hence also satisfiable. The argument is **valid**; it is modus ponens. This says the conclusion follows if the premises hold, not that every premise is factually true.

Related statements. For $p \rightarrow q$, the *converse* is $q \rightarrow p$, the *inverse* is $\neg p \rightarrow \neg q$, and the *contrapositive* is $\neg q \rightarrow \neg p$. Only the contrapositive is always equivalent to the original. “ p only if q ” means $p \rightarrow q$; “ p if q ” means $q \rightarrow p$.

Exam checkpoint. For validity, search for a row with all premises T and conclusion F. One such row disproves validity; otherwise prove there is none.

2 Laws of logic and normal forms

Law	Equivalent expressions
Identity / domination	$p \wedge T \equiv p, p \vee F \equiv p; p \vee T \equiv T, p \wedge F \equiv F$
Idempotent / complement	$p \vee p \equiv p, p \wedge p \equiv p; p \vee \neg p \equiv T, p \wedge \neg p \equiv F$
Double negation	$\neg \neg p \equiv p$
Commutative	$p \vee q \equiv q \vee p; p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r);$ likewise for $\wedge$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q; \neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q) \equiv p; p \wedge (p \vee q) \equiv p$
Implication / biconditional	$p \rightarrow q \equiv \neg p \vee q; p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

A **literal** is a variable or its negation. **DNF** (disjunctive normal form) is an OR of AND-terms of literals. **CNF** (conjunctive normal form) is an AND of OR-clauses of literals. Single literals count as one-literal terms/clauses.

Principal/canonical forms. In PDNF (sum of minterms), each AND-term contains every variable exactly once. In PCNF (product of maxterms), each OR-clause contains every variable exactly once. “Sum/product” here means logical OR/AND, not ordinary arithmetic.

PDNF: OR the minterms of T rows. PCNF: AND the maxterms of F rows.

For a T row, use p when p is T and $\neg p$ when it is F in the minterm. For an F row, *reverse* those choices in the maxterm, so that the clause is false exactly on that row.

Worked question 2: Obtain DNF, CNF, PDNF and PCNF

Question. Write all four forms of $F = p \rightarrow (q \wedge r)$, with variable order p, q, r .

Formula / rule first. $p \rightarrow X \equiv \neg p \vee X$ and $X \vee (Y \wedge Z) \equiv (X \vee Y) \wedge (X \vee Z)$. Canonical terms are selected from the truth-table rows as above.

$$\underbrace{\neg p \vee (q \wedge r)}_{\text{DNF}} \equiv \underbrace{(\neg p \vee q) \wedge (\neg p \vee r)}_{\text{CNF}}.$$

F is true on rows 000, 001, 010, 011, 111 and false on 100, 101, 110, where 1 means T and 0 means F. Thus

$$\begin{aligned} \text{PDNF} &= (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \\ &\quad \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (p \wedge q \wedge r), \\ \text{PCNF} &= (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee r). \end{aligned}$$

The shorter DNF and CNF are not canonical: some terms/clauses omit variables.

Worked question 3: Prove equivalence using laws

Question. Simplify $(p \wedge q) \vee (p \wedge \neg q)$.

Formula / rule first. $(A \wedge B) \vee (A \wedge C) \equiv A \wedge (B \vee C)$; $q \vee \neg q \equiv T$; $p \wedge T \equiv p$.

$$(p \wedge q) \vee (p \wedge \neg q) \equiv p \wedge (q \vee \neg q) \equiv p \wedge T \equiv \boxed{p}.$$

Exam checkpoint. For normal forms: remove $\leftrightarrow$ and $\rightarrow$, push negations inward, then distribute. CNF and DNF need not be unique; principal forms depend on the full declared variable set.

3 Predicates, quantifiers and nested quantifiers

A **predicate** $P(x)$ is a statement depending on a variable from a stated **domain** D . In $\forall x P(x)$ (“for every x ”), the claim must hold for all elements of D . In $\exists x P(x)$ (“there exists an x ”), at least one witness suffices. We use nonempty domains. A variable is **bound** within its quantifier’s scope and **free** otherwise.

$$\boxed{\neg \forall x P(x) \equiv \exists x \neg P(x)} \quad \boxed{\neg \exists x P(x) \equiv \forall x \neg P(x)}.$$

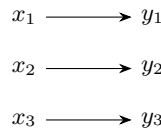
For a finite domain $D = \{d_1, \dots, d_k\}$,

$$\forall x \in D P(x) \equiv \bigwedge_{i=1}^k P(d_i), \quad \exists x \in D P(x) \equiv \bigvee_{i=1}^k P(d_i).$$

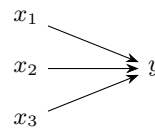
Translation patterns. “All A are B ”: $\forall x(A(x) \rightarrow B(x))$. “Some A is B ”: $\exists x(A(x) \wedge B(x))$. “No A is B ”: $\forall x(A(x) \rightarrow \neg B(x))$. Do not replace the AND in “some” by implication: that could be true merely because the witness is not an A .

Order matters. $\forall x \exists y R(x, y)$ permits a different y for each x ; $\exists y \forall x R(x, y)$ requires one common y . Same-type adjacent quantifiers commute; mixed types generally do not.

$\forall x \exists y$: individual witnesses



$\exists y \forall x$: common witness



Worked question 4: Translate and negate a nested statement

Question. Express and negate “Every student has solved at least one problem.”

Formula / rule first. $\neg \forall x \exists y R(x, y) \equiv \exists x \forall y \neg R(x, y)$. With a single universal domain, restrict “every” by implication and “some” by conjunction.

Let S be the set of students, P the set of problems, and $R(s, p)$ mean “ s solved p .” The statement and its negation are

$$\forall s \in S \exists p \in P R(s, p), \quad \boxed{\exists s \in S \forall p \in P \neg R(s, p)}.$$

The negation says **some student solved no problem**, not “no student solved any problem.” Over a common domain, the original is $\forall s(\text{Student}(s) \rightarrow \exists p(\text{Problem}(p) \wedge R(s, p)))$.

Worked question 5: Decide truth and justify it

Question. Over $\mathbb{Z}$, determine the truth of (a) $\forall x \exists y (y > x)$ and (b) $\exists y \forall x (y > x)$.

Formula / rule first. To prove $\forall x \exists y$, construct y from arbitrary x . To disprove $\exists y \forall x R(x, y)$, prove $\forall y \exists x \neg R(x, y)$.

(a) **True:** for any integer x , choose $y = x + 1$; then $y > x$. (b) **False:** given any proposed y , choose $x = y$. The required $y > x$ becomes $y > y$, which is false.

Exam checkpoint. State the domain. Negate one quantifier at a time, flip $\forall \leftrightarrow \exists$, and negate the predicate. A witness chosen after x may depend on x .

4 Rules of inference, resolution and answer extraction

An **inference rule** licenses a conclusion from specified premises. A proof should name the rule at each non-obvious step.

Rule	Premises	Conclusion
Modus ponens	$p, p \rightarrow q$	q
Modus tollens	$p \rightarrow q, \neg q$	$\neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	$p \rightarrow r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Conjunction / simplification	$p, q / p \wedge q$	$p \wedge q / p$
Addition	p	$p \vee q$
Resolution	$p \vee A, \neg p \vee B$	$A \vee B$

Quantified rules. Universal instantiation: $\forall x P(x) \Rightarrow P(c)$ for a term c in the domain. Existential generalization: $P(c) \Rightarrow \exists x P(x)$. Universal generalization needs an *arbitrary* object, with no special assumptions about it. Existential instantiation introduces a *fresh* witness; do not assume it is an already named object or generalize from it to every object.

Worked question 6: Derive a conclusion; spot an invalid pattern

Question. From $p \rightarrow q, q \rightarrow r, \neg r$, prove $\neg p$. Is $p \rightarrow q, q \therefore p$ valid?

Formula / rule first. Modus tollens: $(A \rightarrow B), \neg B \vdash \neg A$. Invalidity needs a truth assignment with all premises true and conclusion false; $\vdash$ means “is derivable.”

From $q \rightarrow r$ and $\neg r$, infer $\neg q$ by modus tollens. From $p \rightarrow q$ and $\neg q$, infer $\boxed{\neg p}$ by modus tollens. The second argument is **invalid** (affirming the consequent): set $p = \text{F}, q = \text{T}$. Both premises are true, but p is false. Denying the antecedent ($p \rightarrow q, \neg p \therefore \neg q$) is also invalid.

Resolution refutation. Convert premises and the *negation of the desired conclusion* into CNF clauses. Resolve complementary literals until the **empty clause** $\square$ (false) is obtained. This proves no interpretation can make the premises true and conclusion false. In predicate logic, **unification** substitutes terms so that two complementary atoms match. For general quantified inputs, clause conversion may require renaming bound variables and Skolem witnesses/functions; Skolemization preserves satisfiability, not equivalence in the original vocabulary.

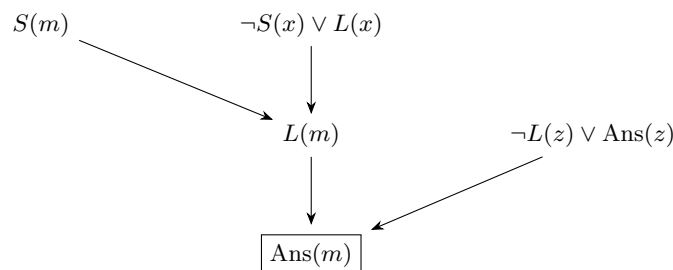
Worked question 7: Resolution and the answer extraction process

Question. All students are learners. Mira is a student. Show that a learner exists and identify one.

Formula / rule first. $\forall x (S(x) \rightarrow L(x))$ yields the clause $\neg S(x) \vee L(x)$. Resolve $A \vee P(x)$ and $B \vee \neg P(t)$ using $\theta = \{x \mapsto t\}$ to obtain $(A \vee B)\theta$.

The facts are $C_1 : \neg S(x) \vee L(x)$ and $C_2 : S(m)$, where m denotes Mira. Resolving with $x = m$ gives $L(m)$. The negation of $\exists z L(z)$ is $\forall z \neg L(z)$, giving $C_3 : \neg L(z) \vee \text{Ans}(z)$. Resolve with $z = m$ to obtain $\square$. Thus $\exists z L(z)$ follows.

Extracting the answer. Track the query variable’s substitution: $z = m$, so **Mira is a witness**. An answer-literal presentation uses the query clause $\neg L(z) \vee \text{Ans}(z)$; resolving it with $L(m)$ derives $\text{Ans}(m)$. This is the resolution-based answer extraction convention: retain the substitution, not merely “true.”



5 Logic methods from the notes: assignments, tables and witnesses

Method A: forced truth values

A satisfying assignment makes every CNF clause true. Begin with a one-literal clause, propagate its forced value, then check every original clause. A failed guess alone does not prove unsatisfiability.

Question. Find a satisfying assignment for $(p \vee q) \wedge (\neg p \vee r) \wedge \neg r$.

Formula / rule first. $X \wedge Y$ is true only when both parts are true; $X \vee F \equiv X$. The clause $\neg r$ forces $r = F$. Then $\neg p \vee r$ forces $p = F$, and $p \vee q$ forces $q = T$. Substitution gives $(F \vee T) \wedge (T \vee F) \wedge T = T$. Thus $(p, q, r) = (F, T, F)$.

Method B: read quantifiers as row/column tests

In a relation table, a 1 means the relation holds. Rows are values of x ; columns are values of y .

$R(x, y)$	$y = a$	$y = b$	$y = c$	Statement	Rule to check	Result
$x = 1$	1	0	0	$\forall x \exists y R(x, y)$	A 1 in every row	T
$x = 2$	0	1	0	$\exists y \forall x R(x, y)$	An all-1 column	F
$x = 3$	1	1	1	$\exists x \forall y R(x, y)$	An all-1 row	T
				$\forall y \exists x R(x, y)$	A 1 in every column	T

Why? Row 3 provides the witness for the third claim; rows 1 and 2 prevent any column from being all 1s. State the table test as your rule before reporting the truth value.

Exactly one versus at least two. On a stated domain, use

$$\exists! x P(x) \equiv \exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x)), \quad \exists x \exists y (P(x) \wedge P(y) \wedge x \neq y).$$

The first includes existence and uniqueness; the second requires distinct witnesses. Without $x \neq y$, both variables may name the same object.

Method C: answer extraction through linked facts

Question. Given $P(A, B)$ and $P(B, C)$, and the rule $\forall x, y, z [P(x, y) \wedge P(y, z) \rightarrow G(x, z)]$, find a supported answer to $G(w, C)$. Here P means “parent of” and G means “grandparent of”; A, B, C are names.

Formula / rule first. Remove implication using $U \rightarrow V \equiv \neg U \vee V$. Resolve complementary atoms only after making them match by one consistent substitution. Carry the query variable in $\text{Ans}(w)$.

Step	Clause	Reason / substitution
1	$\neg P(x, y) \vee \neg P(y, z) \vee G(x, z)$	Rule in clause form
2	$\neg G(w, C) \vee \text{Ans}(w)$	Answer clause for the query
3	$\neg P(x, y) \vee \neg P(y, C) \vee \text{Ans}(x)$	Resolve 1, 2; $z = C, w = x$
4	$\neg P(B, C) \vee \text{Ans}(A)$	Resolve 3 with $P(A, B)$; $x = A, y = B$
5	$\text{Ans}(A)$	Resolve 4 with $P(B, C)$

Thus $w = A$. The shared middle name must match in both parent facts. An answer clause records a witness; ordinary refutation instead uses the negated goal and aims for $\square$.

Preparing existential facts. Replace $\exists y P(y)$ by $P(c)$ with a fresh name. For $\forall x \exists y R(x, y)$ use $R(x, f(x))$, allowing the witness to depend on x . For $\exists y \forall x R(x, y)$ use one fresh c in $R(x, c)$. These are satisfiability-preserving clause preparations, not equivalences in the original vocabulary.

Exam checkpoint. If asked for all supported answers, follow every applicable fact chain. An absent fact is not automatically a false fact. Use a step/reason table whenever it makes a proof easier to audit.

6 Proof techniques: choosing and writing a proof

Technique	What you actually establish
Direct	To prove $P \rightarrow Q$, assume P and derive Q .
Contrapositive	Prove $\neg Q \rightarrow \neg P$, using $P \rightarrow Q \equiv \neg Q \rightarrow \neg P$.
Contradiction	Assume the claim is false and derive an impossibility. For $P \rightarrow Q$, assume $P \wedge \neg Q$.
Cases	Split into exhaustive possibilities and prove the conclusion in every case.
Biconditional	Prove both $P \rightarrow Q$ and $Q \rightarrow P$.
Existence / disproof	Give a witness for $\exists x P(x)$; give a counterexample to disprove $\forall x P(x)$.

Definitions that make number proofs precise. For integers n, k , n is even iff $n = 2k$ for some $k \in \mathbb{Z}$, and odd iff $n = 2k + 1$ for some $k \in \mathbb{Z}$. For integers a, b , $a \mid b$ means $b = ak$ for some $k \in \mathbb{Z}$. A rational number has the form a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$; it can be written with $\gcd(a, b) = 1$.

Worked question 8: Direct proof and contrapositive

Question. (a) Prove the sum of two odd integers is even. (b) Prove that if n^2 is even, then n is even.

Formula / rule first. Odd $n = 2k + 1$; even $n = 2k$, $k \in \mathbb{Z}$. For (b), $P \rightarrow Q \equiv \neg Q \rightarrow \neg P$.

(a) Write the two odd integers as $2a + 1$ and $2b + 1$, where $a, b \in \mathbb{Z}$. Their sum is

$$(2a + 1) + (2b + 1) = 2(a + b + 1).$$

Since $a + b + 1$ is an integer, the sum is even.

(b) Prove the contrapositive: if n is odd, then n^2 is odd. Put $n = 2k + 1$:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

The bracket is an integer, so n^2 is odd. Therefore n^2 even implies n even.

Worked question 9: Contradiction: irrationality of $\sqrt{2}$

Question. Prove that $\sqrt{2}$ is irrational.

Formula / rule first. A rational number equals a/b in lowest terms, with $b \neq 0$. If an integer square is even, the integer is even (Question 8). A contradiction proves the assumed negation false.

Assume $\sqrt{2} = a/b$, with integers a, b coprime. Squaring gives $a^2 = 2b^2$, hence a is even. Let $a = 2k$. Then $4k^2 = 2b^2$, so $b^2 = 2k^2$ and b is even. Both a and b are divisible by 2, contradicting $\gcd(a, b) = 1$. Therefore $\sqrt{2}$ is irrational.

Counterexample and cases. “Every prime is odd” is false because 2 is prime and even. To prove $n(n + 1)$ is even for every integer n , use the rule that consecutive integers have opposite parity: if $n = 2k$, the product is $2k(n + 1)$; if $n = 2k + 1$, it is $2n(k + 1)$. Each is twice an integer.

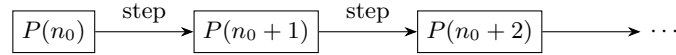
Exam checkpoint. Examples support intuition but do not prove a universal claim. A single counterexample disproves one. In a proof, define the integer parameters and explicitly connect the final expression to the required definition.

7 Mathematical induction and strong induction

Ordinary induction. To establish $P(n)$ for every integer $n \geq n_0$, prove a base case, then the implication $P(k) \rightarrow P(k+1)$ for arbitrary $k \geq n_0$:

$$\boxed{P(n_0) \wedge \forall k \geq n_0 [P(k) \rightarrow P(k+1)] \implies \forall n \geq n_0 P(n).}$$

The **induction hypothesis** is the temporary assumption $P(k)$. It is not permission to assume the result for $k+1$. Strong induction instead assumes *all* $P(n_0), \dots, P(k)$ to prove $P(k+1)$. Both principles are valid; use enough base cases for the step.



Worked question 10: Induction for a sum

Question. Prove $1 + 2 + \dots + n = n(n+1)/2$ for $n \geq 1$.

Formula / rule first. $S_{k+1} = S_k + (k+1)$, where $S_n = \sum_{i=1}^n i$. Apply base case plus $P(k) \rightarrow P(k+1)$.

Base, $n = 1$: $1 = 1(2)/2$. **Hypothesis:** for an arbitrary $k \geq 1$, assume $S_k = k(k+1)/2$. **Step:**

$$S_{k+1} = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

This is exactly the claimed expression with $n = k+1$. Hence the formula holds for all integers $n \geq 1$.

Worked question 11: Induction for divisibility

Question. Prove $3 \mid (4^n - 1)$ for every integer $n \geq 0$.

Formula / rule first. $d \mid m \iff m = dt$ for an integer t ; rewrite $4^{k+1} - 1 = 4(4^k - 1) + 3$ to use the induction hypothesis.

Base: $4^0 - 1 = 0 = 3 \cdot 0$. **Hypothesis:** $4^k - 1 = 3t$ for some integer t . **Step:**

$$4^{k+1} - 1 = 4(4^k - 1) + 3 = 12t + 3 = 3(4t + 1).$$

Since $4t + 1 \in \mathbb{Z}$, the expression is divisible by 3. The result follows by induction.

Worked question 12: Strong induction: factorization into primes

Question. Prove every integer $n \geq 2$ is a product of primes (a prime itself is a one-factor product).

Formula / rule first. Strong induction: assume $P(j)$ for every $2 \leq j \leq k$ to prove $P(k+1)$. A composite positive integer n has a factorization $n = ab$ with $2 \leq a, b < n$.

Base: 2 is prime. **Hypothesis:** every integer from 2 through k is a product of primes. **Step:** if $k+1$ is prime, we are done. Otherwise $k+1 = ab$ with $2 \leq a, b \leq k$. The hypothesis factors both a and b into primes; multiplying those factorizations gives one for $k+1$. Thus the statement holds for every $n \geq 2$.

Method extension: an inequality step. Prove $2^n \geq n+1$ for $n \geq 0$. **Formula / rule first.** Positive multiplication preserves $\geq$; induction requires $P(0)$ and $P(k) \Rightarrow P(k+1)$. Base: $2^0 = 1$. Assume $2^k \geq k+1$. Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k+1) \geq k+2,$$

since $2(k+1) - (k+2) = k \geq 0$. Hence the inequality holds for all $n \geq 0$. The final comparison is the bridge to the next case; show it explicitly.

Exam checkpoint. Always write: starting value, base check, hypothesis with range, calculation/argument for the next value, and final conclusion. A two-step recurrence usually needs two base values.

8 Combinatorics: the pigeonhole principle

The **pigeonhole principle** says that if more than k objects are placed in k boxes, at least one box contains two or more objects. The boxes may represent categories, remainders, dates, or intervals.

Generalized form. For $N \geq 1$ objects and $k \geq 1$ boxes,

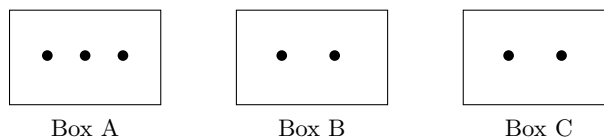
$$\text{some box has at least } \left\lceil \frac{N}{k} \right\rceil \text{ objects.}$$

The ceiling $\lceil x \rceil$ is the least integer at least x . To force at least r objects in a box, the smallest number in the unrestricted setting is

$$N = k(r - 1) + 1.$$

Why? If each box has at most $r - 1$ objects, there are at most $k(r - 1)$ objects altogether. One more is impossible without reaching r . The bound is sharp because $r - 1$ in every box avoids r .

7 objects, 3 boxes: at least $\lceil 7/3 \rceil = 3$ in a box



Worked question 13: Birth months and a minimum guarantee

Question. (a) Among 73 students, how many must share a birth month? (b) How many students guarantee that at least 5 share a birth month?

Formula / rule first. Guaranteed occupancy = $\lceil N/k \rceil$; minimum to force r in one box = $k(r - 1) + 1$. Here the boxes are the $k = 12$ months.

(a) $\lceil 73/12 \rceil = \boxed{7}$. Therefore at least seven students share some month. It need not be a particular month. (b) For $r = 5$,

$$N = 12(5 - 1) + 1 = \boxed{49}.$$

Forty-eight do not suffice: four students in each month avoid five. Forty-nine do suffice by the principle. No assumption about equally likely months is needed.

Worked question 14: Divisibility using remainder boxes

Question. Prove that among any 6 integers, two have a difference divisible by 5.

Formula / rule first. Division algorithm: each integer a has a unique representation $a = 5q + r$ with $q \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4\}$. Equal remainders imply a difference divisible by 5.

Put the six integers in the five boxes indexed by remainder modulo 5. Since $6 > 5$, two integers a, b have the same remainder r . Write $a = 5u + r$ and $b = 5v + r$. Then

$$a - b = 5(u - v), \quad u - v \in \mathbb{Z},$$

so $\boxed{5 \mid (a - b)}$. The integers can be negative; the division algorithm still uses remainders from 0 through 4.

How to recognize a pigeonhole question. It asks to show that some collision or repeated feature *must* occur, without finding its location. Define the objects and boxes before applying a ceiling. If the requested feature is “difference divisible by m ,” remainder classes modulo m are natural boxes.

Exam checkpoint. State the objects, the boxes, their counts, the formula, and what sharing a box means. A guarantee is a worst-case statement; it is not an average or a probability.

9 Pigeonhole methods: partial sums, geometry and hidden boxes

Method D: turn a block sum into a difference

Question. Prove that any list of $m \geq 1$ integers contains a nonempty consecutive block with sum divisible by m .

Formula / rule first. Define $S_0 = 0$ and $S_j = \sum_{t=1}^j a_t$ for $1 \leq j \leq m$. Then

$$S_j - S_i = \sum_{t=i+1}^j a_t \quad (i < j), \quad S_i \equiv S_j \pmod{m} \implies m \mid (S_j - S_i).$$

The $m + 1$ partial sums are the objects; the m residue classes are the boxes. Two partial sums have equal residues. Label their indices $i < j$: their difference is exactly the required block sum. Since $j > i$, the block is nonempty. Include S_0 so that a block starting at the first entry is covered.

Numerical demonstration. For $a = (3, 4, 2, 5)$ and $m = 4$, use the same formula:

j	0	1	2	3	4
S_j	0	3	7	9	14
$S_j \bmod 4$	0	3	3	1	2

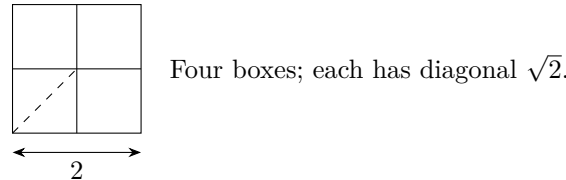
S_1 and S_2 both have residue 3. Thus $S_2 - S_1 = 4 = a_2$, a one-term block divisible by 4. The same proof permits negative entries.

Method E: geometric boxes

Question. Show that among five points in a square of side 2, two are at distance at most $\sqrt{2}$.

Formula / rule first. More than four objects in four boxes forces a pair; distance within a unit square satisfies $d = \sqrt{(\Delta x)^2 + (\Delta y)^2} \leq \sqrt{1^2 + 1^2} = \sqrt{2}$.

Partition into four unit squares. Assign each boundary point to exactly one square by a fixed convention. Five points force two into one square; its diagonal bounds their separation. The conclusion is $d \leq \sqrt{2}$, not $d \leq 1$.



Method F: choose boxes by odd part

Question. Prove that choosing $n + 1$ distinct integers from $\{1, \dots, 2n\}$ forces one selected number to divide another.

Formula / rule first. Every positive integer has a unique form $2^k u$ with u odd and $k \geq 0$. If $a < b$, then $(2^b u) / (2^a u) = 2^{b-a} \in \mathbb{Z}$.

The possible odd parts are $1, 3, \dots, 2n - 1$: exactly n boxes. Two selected numbers share an odd part, say $2^a u, 2^b u$. Distinctness gives $a \neq b$; assume $a < b$. Their ratio is an integer, proving divisibility.

Worst-case capacities. If avoiding the target allows at most c_i objects in box i , then $\sum_{i=1}^k c_i + 1$ forces the target, provided that many objects are available. For a sharp minimum, exhibit an avoiding arrangement with $\sum c_i$ objects. A guarantee about *some* box does not guarantee a result in a specified box.

Exam checkpoint. Choose derived objects when necessary: partial sums for consecutive blocks, small regions for distance, odd parts for divisibility. Explain why equal box labels imply the requested conclusion.

10 Homogeneous linear recurrences with constant coefficients

A **recurrence relation** defines sequence values using earlier ones. We write a_n for the term with index n , usually starting at $n = 0$. An order- d linear recurrence with constant coefficients has the form

$$a_n = c_1 a_{n-1} + \cdots + c_d a_{n-d} + f(n), \quad n \geq d,$$

where the c_i are constants and $c_d \neq 0$. It is **homogeneous** if $f(n) = 0$ and **non-homogeneous** otherwise. The initial values $a_0, \dots, a_{d-1}$ determine a unique sequence. Linearity excludes terms such as a_{n-1}^2 or $a_{n-1}a_{n-2}$; constant coefficients exclude na_{n-1} .

Auxiliary (characteristic) equation. Substitute $a_n = r^n$ in the homogeneous recurrence:

$$r^d - c_1 r^{d-1} - c_2 r^{d-2} - \cdots - c_d = 0.$$

Distinct roots $r_1, \dots, r_d$ give $a_n = \sum_{i=1}^d C_i r_i^n$. A root r of multiplicity m contributes

$$(C_0 + C_1 n + \cdots + C_{m-1} n^{m-1}) r^n.$$

All C_i are constants determined by initial values, not the recurrence coefficients. For real recurrences, a complex conjugate pair $\rho e^{\pm i\theta}$ can be written as $\rho^n (A \cos n\theta + B \sin n\theta)$, with real A, B .

Worked question 15: Distinct characteristic roots

Question. Solve $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, with $a_0 = 2, a_1 = 5$.

Formula / rule first. For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, solve $r^2 - c_1 r - c_2 = 0$; for distinct roots, use $a_n = Ar_1^n + Br_2^n$.

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \implies a_n = A2^n + B3^n.$$

Using $n = 0, 1$ gives $A + B = 2, 2A + 3B = 5$. Subtract twice the first equation from the second: $B = 1$, hence $A = 1$. Thus

$$a_n = 2^n + 3^n \quad (n \geq 0).$$

Check: $a_2 = 4 + 9 = 13$, and the recurrence gives $5(5) - 6(2) = 13$.

Worked question 16: A repeated characteristic root

Question. Solve $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, with $a_0 = 1, a_1 = 6$.

Formula / rule first. A double root r gives $a_n = (A + Bn)r^n$; use both initial values to solve for A, B .

$$r^2 - 4r + 4 = (r - 2)^2 = 0 \implies a_n = (A + Bn)2^n.$$

$a_0 = 1$ gives $A = 1$; $a_1 = 6$ gives $2(1 + B) = 6$, so $B = 2$. Therefore

$$a_n = (1 + 2n)2^n.$$

Check: $a_2 = 5 \cdot 4 = 20$, equal to $4(6) - 4(1) = 20$.

Exam checkpoint. Move every sequence term to one side before reading the characteristic polynomial. Repeated roots require powers of n . Determine every arbitrary constant, then verify the starting values and one further term.

11 Non-homogeneous recurrences: particular solutions

For $a_n - c_1a_{n-1} - \dots - c_da_{n-d} = f(n)$, the general solution is

$$a_n = \text{CF} + \text{PI} = a_n^{(h)} + a_n^{(p)}.$$

CF means complementary function: $a_n^{(h)}$ solves the associated homogeneous equation. **PI** means particular integral: $a_n^{(p)}$ is one particular solution of the full recurrence (no integration is involved). Substitute the trial into the *full* recurrence to determine its coefficients, then impose initial values on the *total* solution.

Forcing term $f(n)$	Particular trial before checking overlap
Constant K	A
Polynomial of degree t	$A_t n^t + \dots + A_1 n + A_0$
$K\lambda^n$, $\lambda \neq 0$	$A\lambda^n$
$P_t(n)\lambda^n$	$\lambda^n(A_t n^t + \dots + A_1 n + A_0)$

Overlap (resonance) rule. If λ is a characteristic root of multiplicity s , multiply the trial by n^s ; otherwise $s = 0$ and no multiplication is needed. A polynomial alone has $\lambda = 1$. Keep all polynomial powers through degree t , including the constant. For sums of forcing terms, add corresponding trials. **Answer order:** auxiliary equation $\rightarrow$ CF $\rightarrow$ PI trial and substitution $\rightarrow$ CF + PI $\rightarrow$ initial values $\rightarrow$ check.

Worked question 17: Polynomial forcing, with no overlap

Question. Solve $a_n = 2a_{n-1} + n$ for $n \geq 1$, with $a_0 = 0$.

Formula / rule first. $a_n = a_n^{(h)} + a_n^{(p)}$; the root is 2, so $a_n^{(h)} = C2^n$. For degree-one forcing, try $a_n^{(p)} = An + B$; $\lambda = 1$ is not a root.

Substitution into $a_n - 2a_{n-1} = n$ gives

$$An + B - 2[A(n-1) + B] = -An + (2A - B) = n.$$

Equating coefficients: $-A = 1$ and $2A - B = 0$, so $A = -1$, $B = -2$. Hence $a_n = C2^n - n - 2$. From $a_0 = 0$, $C = 2$:

$$a_n = 2^{n+1} - n - 2.$$

Check: $a_1 = 4 - 1 - 2 = 1 = 2a_0 + 1$.

Worked question 18: Exponential forcing, with overlap

Question. Solve $a_n = 2a_{n-1} + 2^n$ for $n \geq 1$, with $a_0 = 1$.

Formula / rule first. If the forcing base λ is a simple characteristic root, use $a_n^{(p)} = An\lambda^n$. The total solution is $a_n^{(h)} + a_n^{(p)}$.

The homogeneous root is 2, so a trial $A2^n$ would overlap. Choose $a_n^{(p)} = An2^n$. Substituting gives

$$An2^n - 2A(n-1)2^{n-1} = A2^n = 2^n \implies A = 1.$$

Thus $a_n = (C + n)2^n$. At $n = 0$, $C = 1$, so

$$a_n = (n+1)2^n.$$

Check: $a_1 = 4 = 2(1) + 2$.

Exam checkpoint. Do not use the forcing term in the characteristic equation. First solve the homogeneous part, inspect overlap, choose the full trial, substitute and equate coefficients, then use initial conditions.

12 Auxiliary-equation extensions: higher order and CF + PI

Method G: a third-order equation

Question. Solve $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ ($n \geq 3$), with $a_0 = 3, a_1 = 6, a_2 = 14$.

Formula / rule first. Replace $a_n, a_{n-1}, a_{n-2}, a_{n-3}$ by $m^3, m^2, m, 1$ after moving all terms left. Distinct roots r_i contribute $C_i r_i^n$.

$$m^3 - 6m^2 + 11m - 6 = (m-1)(m-2)(m-3) = 0.$$

Thus $a_n = A + B2^n + C3^n$. The initials give $A + B + C = 3$, $A + 2B + 3C = 6$, and $A + 4B + 9C = 14$. Subtract consecutive equations: $B + 2C = 3$, $2B + 6C = 8$. Subtract twice the first from the second: $2C = 2$. Hence $A = B = C = 1$ and $\boxed{a_n = 1 + 2^n + 3^n}$. Check $a_3 = 36 = 6(14) - 11(6) + 6(3)$.

Method H: a double matching root in the PI

Question. Solve $b_n = 2b_{n-1} - b_{n-2} + 6$ ($n \geq 2$), with $b_0 = 1, b_1 = 4$.

Formula / rule first. $b_n = \text{CF} + \text{PI}$. A root 1 of multiplicity two gives $\text{CF} = A + Bn$. Constant forcing has base 1, so the PI trial is Kn^2 .

The auxiliary equation is $(m-1)^2 = 0$. Substituting the PI into the full recurrence gives

$$K\{n^2 - 2(n-1)^2 + (n-2)^2\} = 6 \implies 2K = 6,$$

so $K = 3$. Combine before using initial values: $b_n = A + Bn + 3n^2$. Then $A = 1$, $4 = 1 + B + 3$, so $B = 0$ and $\boxed{b_n = 1 + 3n^2}$. Check: $b_2 = 13 = 2(4) - 1 + 6$. A trial Kn would still overlap the CF and fail.

Method I: add PIs for different forcing terms

Question. Solve $c_n = 2c_{n-1} + n + 3^n$ ($n \geq 1$), $c_0 = 0$.

Formula / rule first. Linearity allows $\text{PI} = \text{PI}_1 + \text{PI}_2$ for a sum of forcing terms. Use $Kn + L$ for n and $M3^n$ for 3^n ; neither base 1 nor 3 matches the auxiliary root 2.

CF is $A2^n$. For the polynomial trial, $-Kn + (2K - L) = n$, so $K = -1, L = -2$. For the exponential trial, $M3^n - 2M3^{n-1} = 3^n$, so $M = 3$. Therefore

$$c_n = A2^n - n - 2 + 3 \cdot 3^n; \quad c_0 = 0 \Rightarrow A = -1; \quad \boxed{c_n = -2^n - n - 2 + 3^{n+1}}.$$

Check $c_1 = 4 = 2c_0 + 1 + 3$. Add the CF only once.

Complex-root demonstration

Solve $d_n = -d_{n-2}$, $n \geq 2$, $d_0 = 1, d_1 = 0$. **Formula / rule first.** Distinct roots still give $Ar_1^n + Br_2^n$; $i^2 = -1$. The auxiliary equation $m^2 + 1 = 0$ has roots $\pm i$. Thus $d_n = Ai^n + B(-i)^n$. Initial values give $A + B = 1$ and $i(A - B) = 0$, so

$$\boxed{d_n = \frac{i^n + (-i)^n}{2} = \cos(n\pi/2)}; \quad d_0, d_1, d_2, d_3 = 1, 0, -1, 0.$$

Exam checkpoint. Use the actual starting indices. If given a_1, a_2 , substitute $n = 1, 2$ into the total answer. A missing recurrence term means a zero polynomial coefficient; do not shift the remaining powers.

13 Applications of recurrences: building the model

Model before solving. Define what a_n counts and what n measures. Split objects into disjoint cases whose union is exhaustive. State the smallest initial values directly. A recurrence without its index range or starting values is incomplete.

Worked question 19: Binary strings with no consecutive 1s

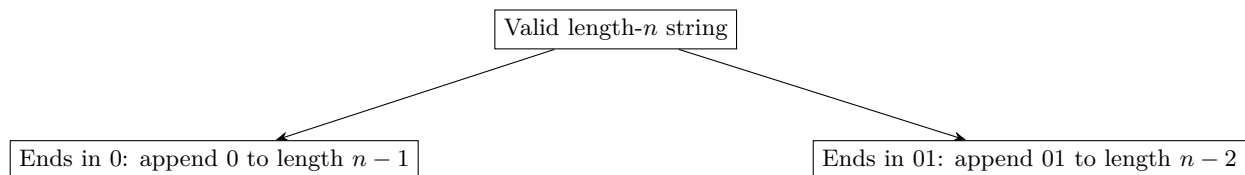
Question. Let b_n count binary strings of length n containing no adjacent 1s. Find a recurrence, starting values, b_5 , and a closed form.

Formula / rule first. Addition principle: disjoint cases add. Characteristic method: $b_n = b_{n-1} + b_{n-2}$ has $r^2 - r - 1 = 0$; distinct roots give a linear combination of their n th powers.

For $n \geq 2$, a valid string either ends in 0, leaving any valid string of length $n - 1$, or ends in 1, which forces the ending 01, leaving any valid string of length $n - 2$. Each construction is reversible, so

$$b_n = b_{n-1} + b_{n-2} \quad (n \geq 2), \quad b_0 = 1, \quad b_1 = 2.$$

There is one empty string; the length-one strings are 0 and 1.



n	0	1	2	3	4	5
b_n	1	2	3	5	8	13

Thus $b_5 = 13$. Put $\phi = (1 + \sqrt{5})/2$ and $\psi = (1 - \sqrt{5})/2$. The roots are ϕ, ψ , so $b_n = A\phi^n + B\psi^n$. From $A + B = 1$ and $A\phi + B\psi = 2$, obtain $A = \phi^2/\sqrt{5}$, $B = -\psi^2/\sqrt{5}$. Therefore

$$b_n = \frac{\phi^{n+2} - \psi^{n+2}}{\sqrt{5}}.$$

Other common models (with their rules).

Model	Definition, recurrence and reason
Stairs: 1 or 2 steps	s_n counts ordered ways to climb n stairs. Last move is 1 or 2: $s_n = s_{n-1} + s_{n-2}$, $n \geq 2$; $s_0 = s_1 = 1$.
Tower of Hanoi	H_n is the minimum moves for n disks in the standard three-peg puzzle. Move $n - 1$ aside, the largest once, then $n - 1$ back: $H_n = 2H_{n-1} + 1$, $n \geq 1$; $H_0 = 0$, yielding $H_n = 2^n - 1$.
Constant growth + deposit	If V_n is a balance after period n , interest rate r per period and deposit d at period end give $V_n = (1 + r)V_{n-1} + d$, $n \geq 1$. For $r \neq 0$, $V_n = (1 + r)^n V_0 + d((1 + r)^n - 1)/r$; for $r = 0$, $V_n = V_0 + nd$.

Regions cut by lines. With no parallel pair and no three concurrent lines, the n th line meets earlier lines at $n - 1$ distinct points, creating n segments and adding n regions. Formula: $R_n = R_{n-1} + n$, $R_0 = 1$, so $R_n = 1 + \sum_{j=1}^n j = 1 + n(n + 1)/2$. For five lines, $R_5 = 16$; five distinct parallel lines instead give six regions.

Exam checkpoint. A recurrence can be correct while the answer is wrong because the initial values belong to a different model. Count the first two or three cases by hand and check that the cases do not overlap.

14 Generating functions: encode, shift and extract

An **ordinary generating function** (OGF) encodes a sequence as coefficients of a formal power series:

$$A(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots, \quad [x^n]A(x) = a_n.$$

Here x is a formal variable, not the sequence index; $[x^n]$ means “coefficient of x^n .” Formal power-series calculations do not require a convergence interval.

Sequence / coefficients	Generating function or identity
$a_n = 1, n \geq 0$	$\sum_{n \geq 0} x^n = \frac{1}{1-x}$
$a_n = c^n, n \geq 0$	$\sum_{n \geq 0} c^n x^n = \frac{1}{1-cx}$
$a_n = n, n \geq 0$	$\sum_{n \geq 0} n x^n = \frac{x}{(1-x)^2}$
$a_n = n+1, n \geq 0$	$\sum_{n \geq 0} (n+1) x^n = \frac{1}{(1-x)^2}$
$a_n = \binom{n+k-1}{k-1}, k \geq 1$	$\frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n$

For integers $u \geq v \geq 0$, $\binom{u}{v} = u!/[v!(u-v)!]$, and $0! = 1$. In particular, $[x^n](1-cx)^{-k} = \binom{n+k-1}{k-1} c^n$.

Shift identities: keep the starting index. For $A(x) = \sum_{n \geq 0} a_n x^n$,

$$\begin{aligned} \sum_{n \geq 1} a_n x^n &= A(x) - a_0, & \sum_{n \geq 2} a_n x^n &= A(x) - a_0 - a_1 x, \\ \sum_{n \geq 1} a_{n-1} x^n &= xA(x), & \sum_{n \geq 2} a_{n-1} x^n &= x[A(x) - a_0], \\ \sum_{n \geq 2} a_{n-2} x^n &= x^2 A(x), & xA'(x) &= \sum_{n \geq 0} n a_n x^n. \end{aligned}$$

Product rule (convolution). If $A(x) = \sum_{n \geq 0} a_n x^n$ and $B(x) = \sum_{n \geq 0} b_n x^n$, then

$$[x^n](A(x)B(x)) = \sum_{j=0}^n a_j b_{n-j}.$$

In counting, a factor lists allowed choices for one component. Multiplication combines choices; exponents add, so extracting x^n selects total size n .

Worked question 20: A counting problem using generating functions

Question. How many nonnegative integer solutions satisfy $u + v + w = 5$, with $u \leq 2$?

Formula / rule first. A variable allowed values $0, \dots, m$ contributes $1 + x + \cdots + x^m$; an unrestricted nonnegative variable contributes $(1-x)^{-1}$. The answer is the coefficient of the required total.

$$G(x) = (1+x+x^2) \frac{1}{1-x} \frac{1}{1-x} = \frac{1-x^3}{(1-x)^3}.$$

Using $[x^n](1-x)^{-3} = \binom{n+2}{2}$,

$$[x^5]G(x) = \binom{7}{2} - \binom{4}{2} = \frac{7 \cdot 6}{2} - \frac{4 \cdot 3}{2} = 21 - 6 = \boxed{15}.$$

Check by fixing $u = 0, 1, 2$: the counts for $v + w = 5, 4, 3$ are 6, 5, 4, respectively; their sum is 15. Here (u, v, w) is ordered, so $(0, 1, 4)$ and $(1, 0, 4)$ are different solutions.

Exam checkpoint. Write what each factor counts. For recurrence sums, do not shift a series without subtracting the initial terms excluded by its lower index.

15 Generating-function methods: restrictions and partial fractions

Restriction dictionary. For a component of size $d \geq 1$ and multiplicity j , the exponent is dj . Multiply one factor for each labelled component type.

Allowed multiplicities	Factor
$j \geq 0$	$1/(1 - x^d)$
$0 \leq j \leq b$	$1 + x^d + \cdots + x^{bd} = (1 - x^{(b+1)d})/(1 - x^d)$
$j \geq a$	$x^{ad}/(1 - x^d)$
$a \leq j \leq b$	$x^{ad}(1 - x^{(b-a+1)d})/(1 - x^d)$

Method J: encode a lower bound as a shift

Question. Count solutions to $u + v + w = 7$, where $0 \leq u \leq 2$, $v \geq 1$, $w \geq 0$ are integers.

Formula / rule first. Use the restriction factors; $[x^n]x^a F(x) = [x^{n-a}]F(x)$ for $n \geq a$ (zero otherwise), and $[x^t](1 - x)^{-2} = t + 1$.

$$G(x) = (1 + x + x^2) \frac{x}{1 - x} \frac{1}{1 - x}, \quad [x^7]G(x) = [x^6] \frac{1 + x + x^2}{(1 - x)^2} = 7 + 6 + 5 = \boxed{18}.$$

The three contributions correspond to $u = 0, 1, 2$. A minimum of one contributes a factor x , not an arbitrary subtraction from the final answer.

Method K: explicitly solve the partial fractions

Question. Solve $a_n = 3a_{n-1} - 2a_{n-2}$ ($n \geq 2$), with $a_0 = 1, a_1 = 4$, using generating functions.

Formula / rule first. $A(x) = \sum_{n \geq 0} a_n x^n$; shifted sums are $A - a_0 - a_1 x$, $x(A - a_0)$ and $x^2 A$. Finally $[x^n](1 - rx)^{-1} = r^n$.

$$A - 1 - 4x = 3x(A - 1) - 2x^2 A \quad \Rightarrow \quad A = \frac{1 + x}{1 - 3x + 2x^2} = \frac{1 + x}{(1 - x)(1 - 2x)}.$$

Set $A = C/(1 - x) + D/(1 - 2x)$. Clear denominators:

$$1 + x = C(1 - 2x) + D(1 - x).$$

Put $x = 1$: $2 = -C$, so $C = -2$. Put $x = 1/2$: $3/2 = D/2$, so $D = 3$. Thus

$$A = \frac{-2}{1 - x} + \frac{3}{1 - 2x} \quad \Rightarrow \quad \boxed{a_n = 3 \cdot 2^n - 2}.$$

Check $a_0 = 1, a_1 = 4$, and $a_2 = 10 = 3(4) - 2(1)$. For a repeated factor $(1 - rx)^2$, include both $C/(1 - rx)$ and $D/(1 - rx)^2$ in the partial-fraction form; coefficients may turn out to be zero.

Ordered pieces versus quantities. If a single positive-size piece has generating function $C(x)$, an ordered list of pieces has

$$1 + C(x) + C(x)^2 + \cdots = \frac{1}{1 - C(x)}.$$

Question. Count total size 3 using pieces of sizes 1 and 2. **Formula / rule first.** Ordered lists use $1/(1 - x - x^2)$; unordered quantities use $1/((1 - x)(1 - x^2))$. Ordered: $(1, 1, 1), (1, 2), (2, 1)$, so the coefficient is 3. Unordered: three 1s or one 1 and one 2, so it is 2. Choose the construction before doing algebra.

Cumulative sums. If $s_n = \sum_{j=0}^n a_j$, convolution with $1 + x + x^2 + \cdots$ gives $S(x) = A(x)/(1 - x)$. This follows because the coefficient at x^n becomes $a_0 + \cdots + a_n$.

Exam checkpoint. Use the method requested in the question. In a generating-function solution, an auxiliary equation alone is insufficient: show the series equation and coefficient extraction.

16 Solving recurrences with generating functions; final review

Method. Define the OGF, multiply the recurrence by x^n , sum over its valid range, use shift identities, solve for the OGF, then extract the coefficient. Partial fractions often convert a rational function into geometric series.

Worked extension of Question 19: derive the OGF and closed form

Question. For $b_0 = 1$, $b_1 = 2$, $b_n = b_{n-1} + b_{n-2}$ ($n \geq 2$), find $B(x)$ and recover b_n .

Formula / rule first. With $B(x) = \sum_{n \geq 0} b_n x^n$, $\sum_{n \geq 2} b_n x^n = B - 1 - 2x$, $\sum_{n \geq 2} b_{n-1} x^n = x(B - 1)$, and $\sum_{n \geq 2} b_{n-2} x^n = x^2 B$.

$$B - 1 - 2x = x(B - 1) + x^2 B \implies \boxed{B(x) = \frac{1+x}{1-x-x^2}}.$$

Let $\phi = (1 + \sqrt{5})/2$, $\psi = (1 - \sqrt{5})/2$. Since $\phi + \psi = 1$ and $\phi\psi = -1$,

$$1 - x - x^2 = (1 - \phi x)(1 - \psi x), \quad B(x) = \frac{\phi^2/\sqrt{5}}{1 - \phi x} - \frac{\psi^2/\sqrt{5}}{1 - \psi x}.$$

The coefficient formula $[x^n](1 - cx)^{-1} = c^n$ now gives

$$\boxed{b_n = \frac{\phi^{n+2} - \psi^{n+2}}{\sqrt{5}}}, \quad B(x) = 1 + 2x + 3x^2 + 5x^3 + 8x^4 + 13x^5 + \dots$$

This independently recovers the characteristic-root answer.

Worked extension of Question 18: a non-homogeneous OGF

Question. Recover a_n when $a_n = 2a_{n-1} + 2^n$ ($n \geq 1$), $a_0 = 1$.

Formula / rule first. $A(x) = \sum_{n \geq 0} a_n x^n$; $\sum_{n \geq 1} a_n x^n = A - 1$; $\sum_{n \geq 1} a_{n-1} x^n = xA$; $\sum_{n \geq 1} 2^n x^n = 2x/(1 - 2x)$.

$$A - 1 = 2xA + \frac{2x}{1 - 2x} \implies A(1 - 2x) = \frac{1}{1 - 2x} \implies A = \frac{1}{(1 - 2x)^2}.$$

Since $[x^n](1 - cx)^{-2} = (n + 1)c^n$, $\boxed{a_n = (n + 1)2^n}$, as before.

Last-pass exam checklist

If the question asks...	Write these steps explicitly
Truth / validity	Declare variables; use 2^k rows; inspect the final column or a countermodel.
Normal forms	Eliminate arrows; use De Morgan; distribute; use all variables for principal forms.
Quantifiers	State domain and predicate; preserve order; flip every quantifier when negating.
Inference / extraction	Name rules; negate the target for refutation; retain substitutions for witnesses.
Proof / induction	State method; use definitions; show base(s), hypothesis and step for induction.
Pigeonhole	Identify objects and boxes; use $\lceil N/k \rceil$ or $k(r - 1) + 1$; interpret the result.
Recurrence	State index range and initials; solve characteristic equation; check multiplicities and forcing overlap.
Generating function	Define the series; show shifted sums; solve algebraically; extract coefficients.

Exam checkpoint. Write the applicable formula or logical rule before substitution. For a proof, show why each step follows. For a numerical answer, include one check. Draw diagrams only when they clarify the structure; label all symbols.